

B Inquiring and designing

- i. **explain** a problem or question to be tested by a scientific investigation
- ii. **formulate** a testable hypothesis and explain it using scientific reasoning
- iii. **explain** how to manipulate the variables, and explain how data will be collected
- iv. **design** scientific investigations.

Command Term	Meaning	Student-Friendly Meaning
Explain	Give a detailed account of the reasons or causes of something.	Tell how or why something happens. Give reasons and details.
Formulate	Create or develop something in a clear and logical way.	Create or write your own idea, question, or hypothesis.
Design	Produce a plan, method, or solution for a specific purpose.	Plan how you will do something step by step.

Examples in Science

Explain

- Explain why plants need sunlight.
- *Plants need sunlight because they use it to make food through photosynthesis.*

Formulate

- Formulate a hypothesis.
- *If a plant gets more sunlight, then it will grow taller because sunlight helps plants make food.*

Design

- Design an investigation to test plant growth.
- *Plan the materials, variables, method, and data table needed for the experiment.*

Easy Way to Remember

- **Explain = Tell why**
- **Formulate = Create**
- **Design = Plan**

Explain a problem/question → Tell what you want to find out and why it is important.

Formulate a hypothesis → Create a prediction with a scientific reason.

Design an investigation → Plan a fair test to answer the question.

Strand	What it means	Easy understanding	Formula
Explain a problem or question to be tested by a scientific investigation	State the question you want to investigate.	What do I want to find out? Write a clear research question that can be tested.	Testable Should have dependent and independent variable Starting Prompts-How? To what extent?
Formulate a testable hypothesis and explain it using scientific reasoning	Make a prediction and give a reason.	What do I think will happen and why?	Write an "If...then...because ..." statement.

Explain how to manipulate the variables, and explain how data will be collected	Describe what you will change, measure, and keep the same. Explain how you will record observations.	Independent variable-What will I change? Dependent Variable- What will I measure? Control Variable-What will stay the same? How will I collect my results?	
Design scientific investigations	Plan a fair test to answer the research question.	How will I carry out the investigation step by step?	Materials Method = Steps I will follow Data = Information I collect Investigation = A fair test to answer the question

Example

Research Question:

How does the amount of sunlight affect plant growth?

Hypothesis:

If a plant receives more sunlight, then it will grow taller because plants need sunlight to make food.

Variables:

- Independent Variable: Amount of sunlight
- Dependent Variable: Plant height
- Controlled Variables: Type of plant, soil, water, pot size

Data Collection:

Measure the height of the plant every day and record it in a table.

Investigation Design:

1. Take two identical plants.
2. Place one in full sunlight and one in partial sunlight.
3. Water both plants equally.
4. Measure their height daily for one week.
5. Record the results in a table and compare them.

Classroom Practice /Home task Activity

Scenario: A toy car rolls down ramps of different heights.

1. **Research Question:** _____
2. **Hypothesis:** If _____, then _____, because _____.
3. **Independent Variable:** _____
4. **Dependent Variable:** _____
5. **Controlled Variables:** _____

Scenarios

- Does a longer pencil roll farther than a shorter pencil?
- Does more water make a paper boat sink faster?
- Does the size of an ice cube affect how long it takes to melt?
- Does the amount of air in a balloon affect how far it travels?
- Does the distance from a fan affect how fast a paper spinner falls?

students can easily identify:

- **Research Question = What am I finding out?**
- **Hypothesis = What do I predict?**
- **Independent Variable = What do I change?**
- **Dependent Variable = What do I measure?**
- **Controlled Variables = What do I keep the same?**